Ethereum Wallet Behaviour Segmentation

Detailed final presentation: data pipeline, role discovery, validation evidence, stability tests, and findings

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2.74M
wallets processed

6
interpretable wallet roles

0.932
network assortativity

8.13M
wallet-month transition pairs

Final Project Presentation

What question does the project answer?

Research Question

Can Ethereum wallets be organized into behavioural roles that are meaningful beyond a clustering algorithm?

Meaningful means that roles should be interpretable, visible in the transaction network, and reasonably stable across time and feature choices.

Why it matters

- Blockchain data is public but difficult to interpret at wallet scale.
- Many studies describe activity, but fewer validate behavioural roles across multiple evidence types.
- A simulator or risk model needs realistic agent types, not assumed personas.

The focus is not only the final segmentation, but how each result was tested and validated.

PROJECT EVOLUTION

From simulator idea to empirical wallet roles

Initial approach

AI-governed crypto-bank simulator

- Simulated depositors, borrowers, treasury, and loan pool.
- AI-assisted credit scoring, risk analysis, governance, and explanations.
- Goal was an application prototype with interpretable decisions.

Final direction

Real Ethereum behaviour first

- Instead of assuming agents, discover wallet types from transaction data.
- Validate the roles with network evidence, statistics, and time.
- Use the empirical roles as a stronger foundation for future simulation.

The pivot made the project more academic: the deliverable became a defensible methodology, not only a prototype.

METHOD

Completed analysis pipeline

1 **Transaction data**

Read daily Ethereum transaction records and isolate wallet-level activity.

2 **Feature engineering**

Convert raw transactions into activity, value-flow, directionality, volatility, and concentration features.

3 **Role discovery**

Run KMeans clustering and interpret six wallet roles from normalized behavioural profiles.

4 **Network validation**

Test whether feature-based roles also organize transaction-network structure.

5 **Temporal analysis**

Assign monthly wallet states and measure role retention and transition pathways.

6 **Robustness checks**

Add correspondence, permutation, feature-set, and observation-density tests.

Output: a validated behavioural segmentation framework for Ethereum wallets.

From raw transactions to wallet-level behaviour

83.5M
daily rows read

46.2M
matched rows used

2.74M
wallets in final output

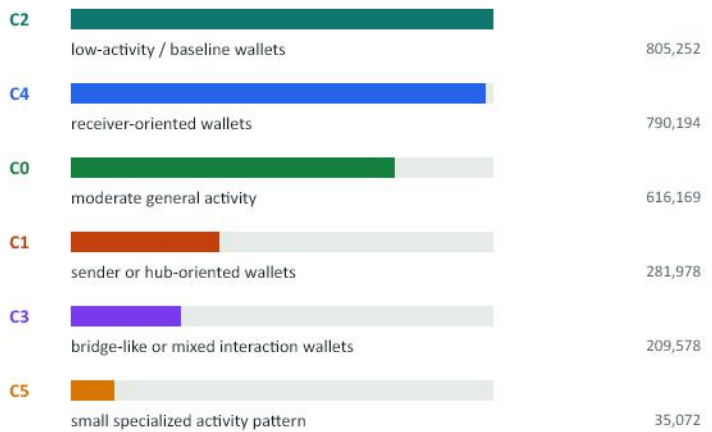
33
numeric behavioural features

Feature families

- Activity intensity: how often wallets transact and how persistent their activity is.
- Value flow: how much value wallets send and receive.
- Directionality: whether behaviour is sender-oriented, receiver-oriented, or mixed.
- Volatility and concentration: whether activity is steady, bursty, broad, or focused.

Finding from this stage: the dataset was large enough to support wallet-level modelling, but sparse enough that stability had to be tested explicitly.

Six behavioural wallet roles

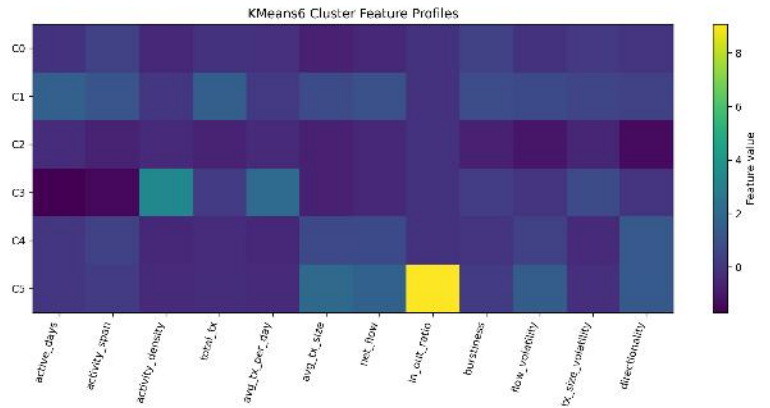


Interpretation principle

The labels were not manually assigned before clustering. They were interpreted after comparing cluster feature profiles, network position, and temporal behaviour.

This gives the project a behavioural vocabulary for millions of addresses.

What separates the roles?



Reading the heatmap

- Rows are wallet roles.
- Columns are normalized behavioural features.
- Color differences show which behaviours distinguish each role.
- The clusters become explainable profiles rather than black-box labels.

Do the roles organize the transaction graph?

23,755	80,596	0.777	0.932
nodes in network summary	transaction edges	modularity	assortativity

Modularity, simply

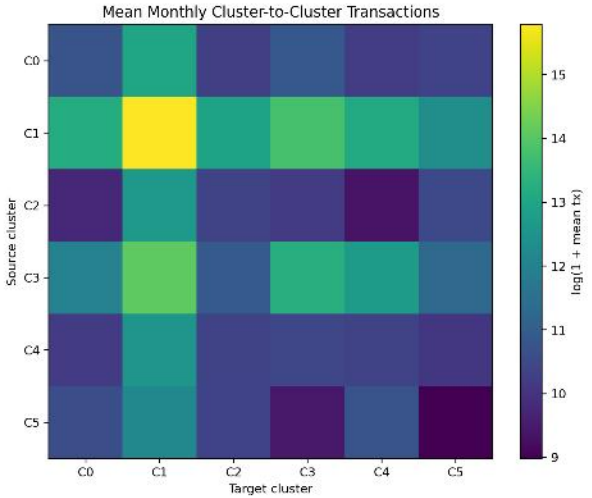
If wallets in the same role form clear communities in the network, modularity is high. Here it suggests strong community structure.

Assortativity, simply

If wallets connect more often to wallets with the same role, assortativity is high. Here it shows strong role-to-role organization.

- Random-label comparison: real same-role structure was about 5.66 times stronger than random.
- Empirical p-value: 0.000999.

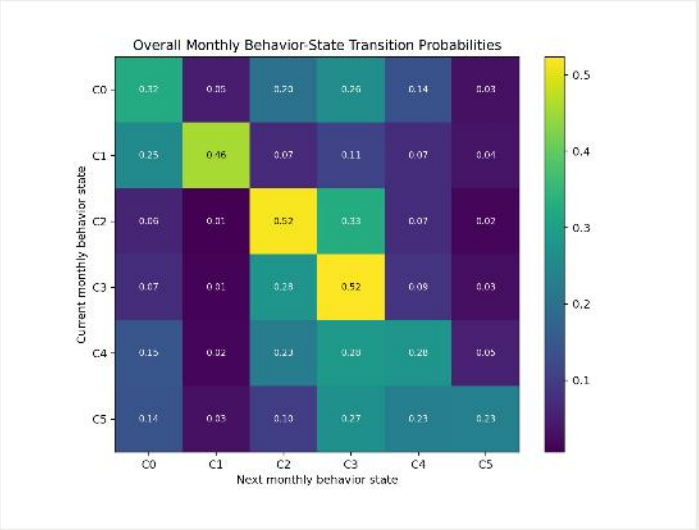
How wallet roles transact with each other

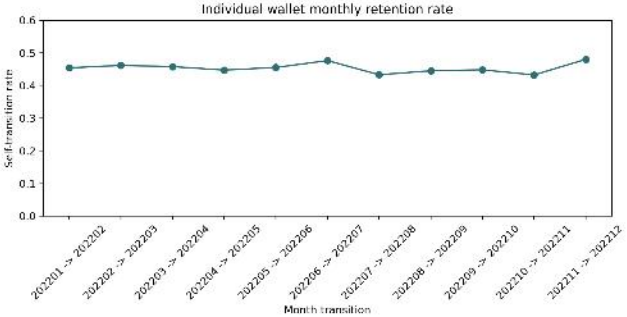


Main reading

- Transactions are not evenly distributed across roles.
- Some roles interact strongly inside their own category.
- Hub-like and bridge-like roles help connect otherwise different wallet populations.
- This supports the idea that roles are part of the network structure, not only feature-space clusters.

Wallet behaviour changes, but not randomly





- 11 month-to-month transitions; 8,130,751 consecutive wallet-month pairs.
- Mean monthly self-transition rate: 0.4538, with monthly values from 0.4324 to 0.4802.
- C3 was the most stable state; C5 was the least stable.

FINDINGS MAP

What evidence supports the six-role result?

Feature evidence

The heatmap showed distinct normalized behavioural profiles. This gave each cluster an interpretable role description instead of only a numeric label.

Statistical evidence

Non-parametric tests and Cliff's delta showed that major cluster differences were not only statistically significant, but also practically large.

Network evidence

The same labels produced high modularity, high assortativity, and much stronger same-role interaction than random labels.

Temporal evidence

Monthly transition matrices showed moderate retention, common transition pathways, and state-specific differences in persistence.

Robustness evidence

Additional checks tested correspondence across analysis variants, feature-set redundancy, permutation significance, and observation density.

Interpretation evidence

Internal subcluster analysis suggested that the six-role solution is a useful top-level vocabulary, with finer patterns inside some roles.

This is the central logic of the project: discover roles first, then test them from several independent angles.

Why the six-cluster solution was kept

Why KMeans6

- It produced roles that were understandable from the feature profiles.
- It aligned well with the transaction-network validation results.
- It gave a practical top-level role vocabulary for millions of wallets.
- It connected naturally to the original simulator goal: realistic agent types.

What we checked

- Alternative-k diagnostics suggested that more clusters can reveal additional fine-grained structure.
- Internal subcluster tests showed that some main roles contain meaningful subtypes.
- Because the goal was an interpretable role layer, we treated KMeans6 as the main segmentation and subclusters as secondary detail.

Finding: six clusters are not the only possible partition, but they are a defensible high-level behavioural layer.

STATISTICAL VALIDATION

Are the clusters actually separated?

Tests used

- Kruskal-Wallis tests to check whether feature distributions differ across clusters.
- Mann-Whitney pairwise comparisons for cluster-to-cluster differences.
- Cliff's delta to measure practical effect size, not only statistical significance.

Main finding

The strongest separated pair was C1 versus C2 across 33 features, with mean absolute Cliff's delta around 0.741 and maximum around 0.990.

This means the clusters were not just numerically different; several differences were large enough to be behaviourally meaningful.

Methodological implication: we used both significance and effect size so the claims do not depend only on large sample size.

Checking feature clusters against communities

What we built

- A transaction network where nodes are wallets and edges represent observed transfers.
- A role-level network summary to study how clusters connect.
- Random-label baselines to test whether the same pattern appears by chance.

Network metrics

23,755 nodes
80,596 edges
0.777 modularity
0.932 assortativity

High modularity means clear communities. High assortativity means wallets tend to connect to wallets with similar roles.

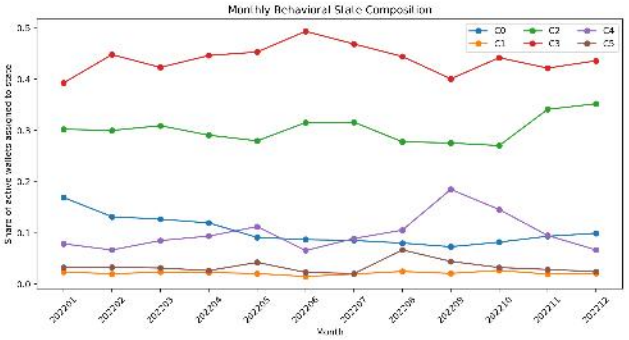
Random-label result

The real labels produced much stronger same-role structure than random labels.

Same-role structure was about 5.66 times stronger than the random comparison, with p = 0.000999.

Finding: the behavioural roles are visible in the transaction graph, not only in the engineered feature table.

What we found from monthly state transitions



Temporal findings

- Wallets do move between roles, so roles are behavioural states rather than permanent identities.
- The system is structured: mean self-transition was 0.4538 across 11 month-to-month transitions.
- C3 showed the strongest retention at 0.5237, while C5 was the least stable at 0.2340.
- Off-diagonal transitions show likely pathways between low-activity, receiver, hub, and bridge-like behaviour.

Finding: stability is moderate, not absolute. That is useful because it captures both persistent roles and behavioural change.

Did similar roles appear across analysis variants?

Permutation test summary

Observed overlap	Random null mean
0.3952	0.2539

Difference	Empirical p-value
+0.1413	0.000999

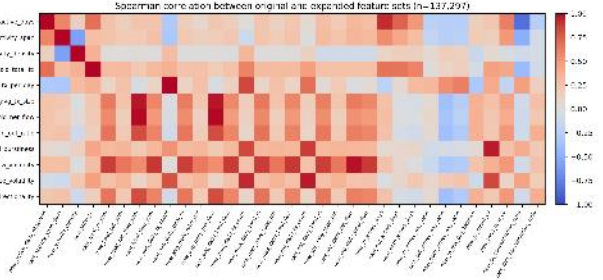
Interpretation: alignment between roles across analyses was stronger than expected from randomized labels.

Formal matching rule

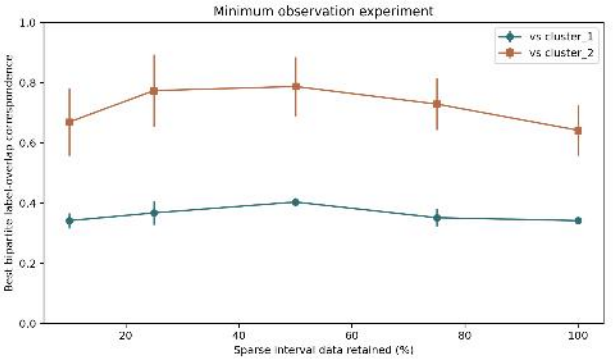
- We compared cluster profiles using four metrics: cosine, weighted Euclidean, Mahalanobis, and Wasserstein.
- Hungarian matching was used to align clusters rather than comparing labels by name.
- Five cluster pairs appeared as core correspondences under at least three of four metrics.
- Two pairs appeared under all four metrics: 2->2 and 4->1.

Finding: role structure is partly reproducible across analysis variants, but label-overlap and profile-distance correspondence are separate results.

Are the findings caused by duplicate features or too much data?



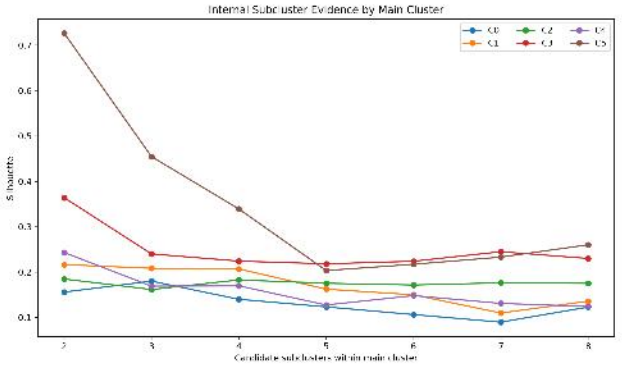
Feature-set check: sampled 137,297 overlapping wallets. Median absolute Spearman correlation was 0.2612; only 8.08 percent of pairs were at least 0.8.



Observation-density check: repeated KMeans on 10, 25, 50, 75, and 100 percent subsamples showed recoverable core structure, but should not be overclaimed as a universal threshold.

- Finding 1: expanded features are related to the original features, but not redundant.
- Finding 2: core roles can reappear under reduced observation, while rare or specialized roles are more data-sensitive.

The six roles are the top layer, not the whole story



What this showed

- We tested whether each main role contains finer behavioural subtypes.
- C5 had the strongest internal subcluster signal, with silhouette around 0.726.
- C3 also showed meaningful bridge-like internal structure.
- This suggests KMeans6 is useful as a high-level role layer, but some roles contain important sub-roles.

Finding: the final system can support both a coarse role vocabulary and finer agent-type modelling later.

WHAT WE ACCOMPLISHED

Chronological summary of the project work

ENS 491

- Started with an AI-governed crypto-bank simulator concept.
- Defined motivation around interpretable blockchain finance behaviour.
- Identified that realistic agent types were missing.

Core build

- Built wallet-level feature pipeline from raw Ethereum transactions.
- Clustered 2.74M wallets into six interpretable roles.
- Interpreted roles using feature profiles and cluster sizes.

Final stage

- Validated roles with graph structure, statistical separation, and temporal transitions.
- Added correspondence, permutation, feature-redundancy, and observation-density checks.
- Added feature redundancy and observation-density experiments.

The project moved from engineering a simulator to academically validating the behavioural foundation such a simulator would need.

FINAL INTERPRETATION

What we can claim, and what we should be careful about

Strong claims supported

- Wallet roles are interpretable from engineered behavioural features.
- Roles align strongly with transaction-network structure.
- Wallet behaviour has moderate temporal persistence.
- Cross-analysis checks show non-random role correspondence.

Careful wording needed

- Roles are behavioural states, not verified real-world identities.
- Subsampling results show recovery in this dataset, not a universal minimum-data threshold.
- Distance-metric correspondence and wallet-label overlap are different validation results.
- Future work should replicate across more time windows and external datasets.

Final point: the academic contribution is validated role discovery for Ethereum wallets, with evidence from features, graph structure, time, and robustness checks.